

Source Code - Sales Performance Dashboard

The following source code implements the Sales Performance Dashboard using Python and Pandas. It covers dataset loading, data inspection, cleaning, transformation, feature engineering, KPI calculation, and sales aggregation.

Source Code:

```
import pandas as pd
import numpy as np

# Load Dataset
df = pd.read_csv("Sample - Superstore.csv")

print("Dataset Shape:", df.shape)
print(df.head())
print(df.info())

# Data Inspection
print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Records:", df.duplicated().sum())

# Remove Duplicate Records
df = df.drop_duplicates()

# Standardize Column Names
df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
)

# Convert Date Columns
df["order_date"] = pd.to_datetime(df["order_date"])
df["ship_date"] = pd.to_datetime(df["ship_date"])

# Feature Engineering
df["order_year"] = df["order_date"].dt.year
df["order_month"] = df["order_date"].dt.month
```

```
df["order_quarter"] = df["order_date"].dt.quarter
```

```
# Calculate Delivery Days
```

```
df["delivery_days"] = (  
    df["ship_date"] - df["order_date"]  
) .dt.days
```

```
# Calculate Profit Margin
```

```
df["profit_margin"] = (  
    df["profit"] / df["sales"]  
) * 100
```

```
# Calculate Sales Per Unit
```

```
df["sales_per_unit"] = (  
    df["sales"] / df["quantity"]  
)
```

```
# KPI Calculation
```

```
total_sales = df["sales"].sum()  
total_profit = df["profit"].sum()  
total_quantity = df["quantity"].sum()  
total_orders = df["order_id"].nunique()
```

```
profit_margin = (  
    total_profit / total_sales  
) * 100
```

```
# Sales by Category
```

```
category_sales = (  
    df.groupby("category")["sales"]  
    .sum()  
    .sort_values(ascending=False)  
)
```

```
# Sales by Region
```

```
region_sales = (  
    df.groupby("region")["sales"]  
    .sum()  
    .sort_values(ascending=False)  
)
```

```
# Monthly Sales
```

```
monthly_sales = (  
    df.groupby("month")["sales"]  
    .sum()  
    .sort_values(ascending=False)  
)
```

```
df.groupby("order_month")["sales"]
.sum()
)
```

```
# Display KPI Results
```

```
print("\nTotal Sales:", total_sales)
print("Total Profit:", total_profit)
print("Total Quantity:", total_quantity)
print("Total Orders:", total_orders)
print("Profit Margin:", profit_margin)
```

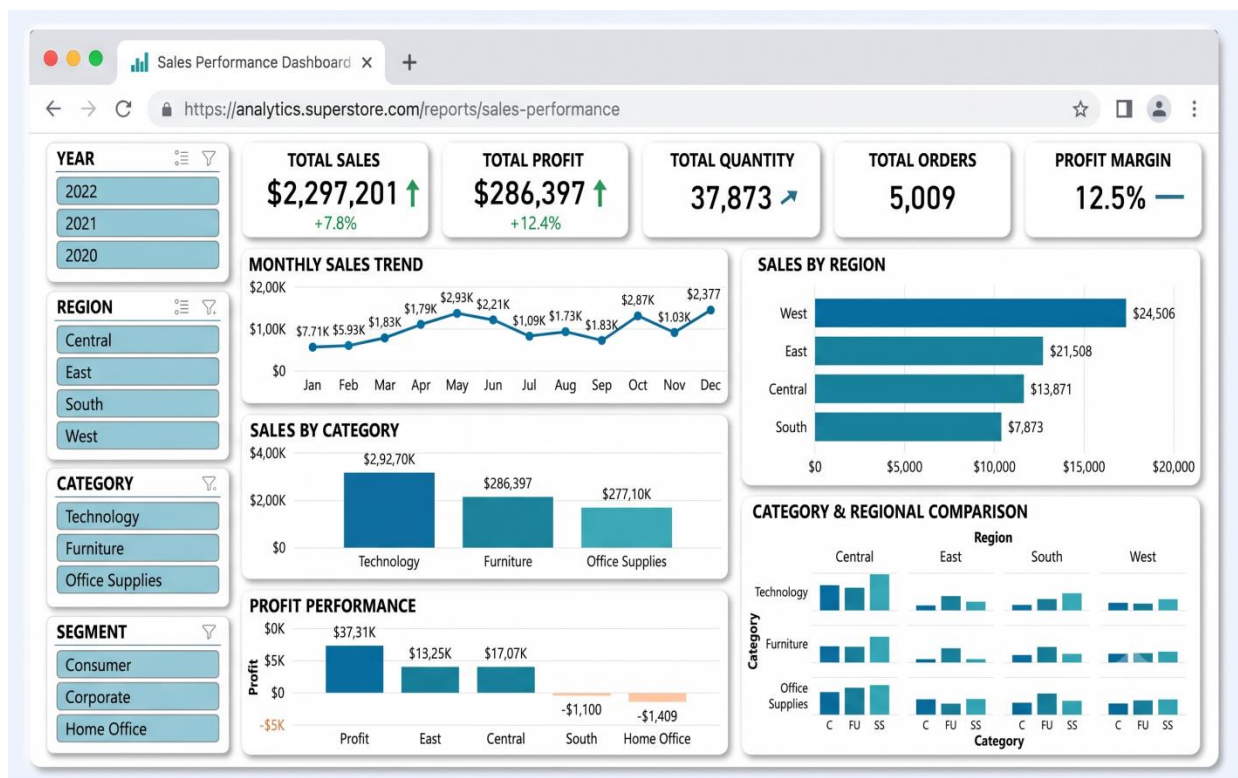
```
# Display Analysis Results
```

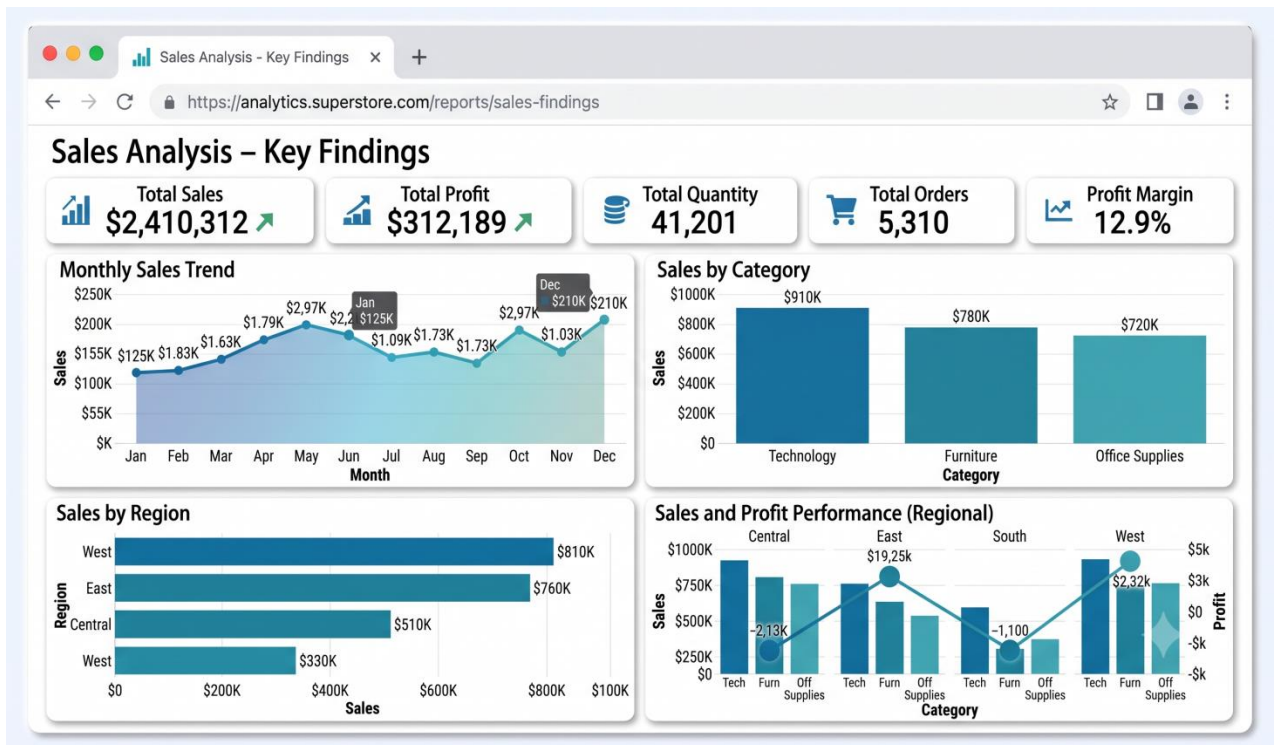
```
print("\nSales by Category:")
print(category_sales)
```

```
print("\nSales by Region:")
print(region_sales)
```

```
print("\nMonthly Sales : ")
print(monthly_sales)
```

Outputs :





Implementation Summary:

The implementation performs data cleaning, date conversion, feature engineering, KPI calculation, and aggregation of sales data. The resulting information is used for category-wise, region-wise, and monthly sales performance analysis and dashboard visualization. The implementation follows the algorithm and methodology described in the project.